



KARNATAKA STATE COUNCIL FOR SCIENCE AND TECHNOLOGY
Indian Institute of Science campus, Bengaluru

Telephone: 080 -23600978, 23341652 || Email: spp@kscst.org.in
 Website: www.kscst.org.in/spp.html or https://kscst.karnataka.gov.in/en

**FORMAT FOR STUDENT PROJECT PROPOSAL FOR THE
 48th SERIES OF STUDENT PROJECT PROGRAMME**

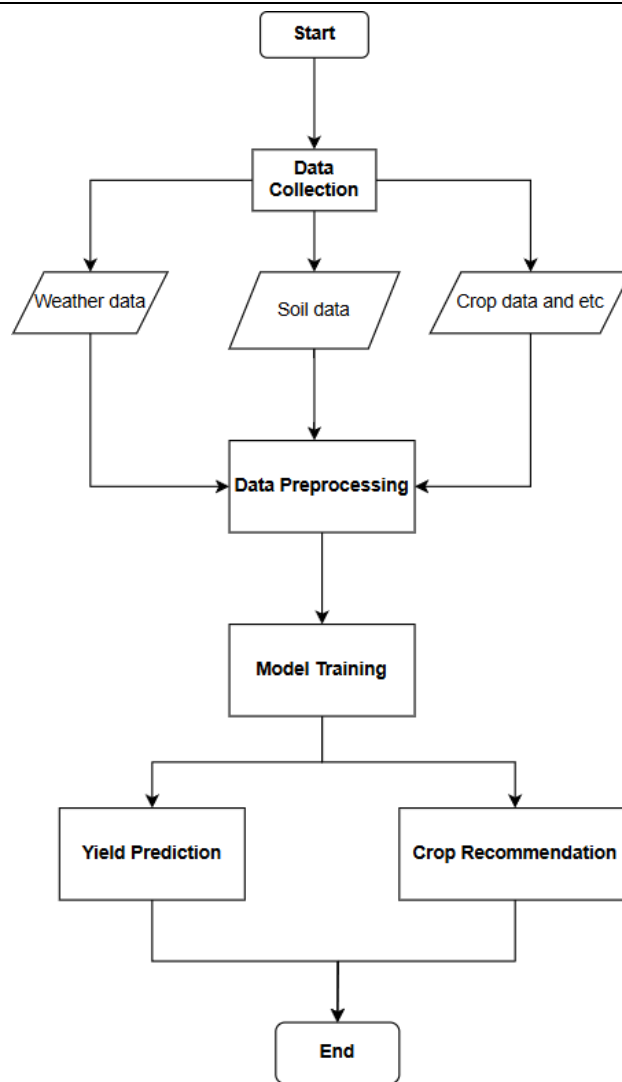
(Handwritten proposals will not be accepted, please fill all the details in this MS word file, insert images / diagrams wherever necessary. Convert to pdf file, get it approved from the project guide / head of the department and principal of your institution. Keep ready the scanned pdf file of 1) Declaration and Endorsement 2) details of processing fees made and fill-up the Google Form.

<https://forms.gle/ks2WxWB4ei1hgv9D9>

1.	Name of the College: Jawaharlal Nehru New College of Engineering
2.	Project Title: "AGRIVISION:AI-ENHANCED YIELD FORECASTING AND SMART CROP SELECTION"
3.	Branch: Computer Science and Engineering
4.	Theme (as per KSCST poster): 6. ARTIFICIAL INTELLIGENCE (agriculture) (The project proposals shall mandatorily be from one of the broad themes / areas. Visit website www.kscst.org.in/spp.html)
5.	Name(s) of project guide(s): 1. Name: Prof. Ayesha Siddiqa Email id: ayeshasiddiqa@jnnce.ac.in Contact No :9742153106
6.	Name of Team Members (Strictly not more than four students in a batch): (Type names in Capital Letters as provided in your college) (Please paste the latest passport size photograph adjacent to your respective names) Name: Umme Hani USN No.: 4JN21CS178 Email id: ummehani181003@gmail.com Mobile No: 6363003776

	Name: Shambhavi S V USN No.: 4JN22CS415 Email id: shambhavisv13@gmail.com Mobile No:6362016508 Name: Shreya G USN No.:4JN21CS153 Email id: kskrupa43@gmail.com Mobile No:7676438289 Name: Varshaa S K USN No.: 4JN21CS180 Email id: varshaavarsh@gmail.com Mobile No:9449372323
7.	Team Leader of the Project: Name: Umme Hani USN No.:4JN21CS178 Email id: ummehani181003@gmail.com Mobile No: 6363003776
8.	Processing Fee Details (Through Online Payment only): (processing fee of Rs. 1180/-) Please furnish the payment details in the format provided in the last page of the proposal.
9.	Date of commencement of the Project: 9-09-2024
10.	Probable date of completion of the project: 31-01-2025

11.	Scope / Objectives of the project: 1. Data Collection : It is to gather real-time data on soil, weather, and crop conditions. This helps provide farmers with accurate crop predictions and smart recommendations for better yields. 2. Data Preprocessing: Clean the input data to remove any inconsistencies. Normalize and transform the data for effective model training. 3. Model Training: Develop and train the AI model using the preprocessed data. Validate the model to ensure high accuracy and reliability. 4. Yield Prediction: Use the trained model to estimate the expected crop yield. Provide yield forecast based on historical trends and environmental factors. 5. Crop Recommendation: Analyze the data to suggest the most suitable crops for current conditions. Help optimize crop choices to maximize productivity and profitability.
12.	Methodology: For this project, we'll gather data on weather, soil quality, and past crop yields using online sources and available databases. With this information, we'll train AI models to predict how much crops will yield and suggest the best crops for each region, considering environmental and market factors. It provide the real-time updates, so farmers can make quick adjustments, leading to smarter and more sustainable farming choices.



13. Expected Outcome of the project:

- 1. Better Harvest Predictions:** Farmers will have a clearer idea of their expected harvest, making it easier to plan and feel confident about the season ahead.
- 2. Smarter Resource Use:** Farmers will be able to use water, fertilizer, and other resources more efficiently, saving money and reducing waste.
- 3. Real-Time Insights:** With updates from AI models, farmers can monitor soil and weather conditions as they happen, so they can make quick, informed adjustments.
- 4. Lower Risk of Crop Loss:** By selecting crops that suit their land and keeping an eye on conditions, farmers can reduce the chances of crop failure.

	5. Higher Yields and Profits: With better predictions and choices, farmers are likely to see stronger harvests and better earnings. 6. Sustainable Farming Practices: Using resources carefully and cutting down on waste supports a more sustainable approach to farming, benefiting both the farmer and the environment.						
14.	Is the project proposed relevant to the Industry / Society or Institution? Yes / No: Yes If yes, please provide details of the industry / institution and contact details: This project is highly relevant to the agricultural industry as it leverages AI to improve crop yield prediction and optimize resource use, benefiting both Agri Tech companies and farmers. It also supports research institutions dedicated to advancing farming practices with technology. By enhancing productivity, reducing waste, and promoting sustainable agriculture, this project has the potential to make a significant impact on both the industry and academic research.						
15.	Can the product or process developed in the project be taken up for filing a Patent? Yes / No: If all the outcomes expected holds good then we may apply for the patent Prior Art search done? Yes/No: No Note: If your answer is “Yes”, you may contact Patent Information Centre of KSCST. For more details, email: pic@kscst.org.in						
16.	Budget details (break-up details should be given): Note: KSCST will provide nominal grant support for carrying out the project by students if selected by the project selection committee. <table border="1"> <thead> <tr> <th>Budget</th><th>Amount</th></tr> </thead> <tbody> <tr> <td>a) Materials / Consumables (Google collab, Weather API, kaggle, MySQL, data source, ml model , visualization, PowerBI ,software (python))</td><td>2000.00</td></tr> <tr> <td>b) Labor (Assistance for soil collection, preprocessing and integration support, data cleaning/labelling)</td><td>4200.00</td></tr> </tbody> </table>	Budget	Amount	a) Materials / Consumables (Google collab, Weather API, kaggle, MySQL, data source, ml model , visualization, PowerBI ,software (python))	2000.00	b) Labor (Assistance for soil collection, preprocessing and integration support, data cleaning/labelling)	4200.00
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Total	15900							
17.	Any other technical details (Please specify): "AgriVision: AI-Enhanced Yield Forecasting and Smart Crop Selection" and AI algorithms (Dynamic Programming, SVM) for real-time data analysis. It uses web dashboard to provide accurate yield predictions and crop recommendations, enhancing decision-making and promoting sustainable agriculture.							
18.	SPP Coordinator (Identified by the college): Note: To be identified by the principal of the institution. The project proposals must be submitted to KSCST through SPP coordinator designated by the principal. Name: Prof. / Dr. / Mr. / Mrs. Email id: Dr. Chethan S G Contact No.:8095040747							

Name of the Project Guide: Prof. AyeshaSiddiqua
Email id:ayeshasiddiqua@jnnce.ac.in
Contact No :9742153106

Name of the HOD: Dr. Jalesh Kumar
Email id: hod_cse@jnnce.ac.in
Contact No.: 9449423361

DECLARATION

(From Project Students)

(To scan this page and enclose in the project proposal)

We, the project team hereby declare that the details enclosed in the project proposal (Title of the Project: AGRIVISION:AI-ENHANCED YIELD FORECASTING AND SMART CROP SELECTION", Branch: Computer Science And Engineering, College: Jawaharlal Nehru New College Of Engineering Shimogga) are true and correct to the best of our knowledge and belief. We undertake to inform the Karnataka State Council for Science and Technology (KSCST) of any changes to the project title or team members' names immediately through our project guide or the SPP Coordinator of our institution.

Additionally, we declare that the project work is original and not a result of copying or purchasing. We are committed to completing the project independently, with support from our faculty and project guide, while utilizing the facilities provided by the college. We will not engage in plagiarism and pledge to be sincere and dedicated in executing and completing the project as proposed.

We understand that if any of the above information is found to be false, untrue, or misleading, we may be held liable. We authorize the sharing of the project information contained in this proposal with KSCST, Bengaluru.

We acknowledge that the project team must exhibit and demonstrate the project, participate in the mid-term evaluation of sanctioned projects, and engage with experts. Additionally, we must exhibit the project at the Annual State-Level Poster Presentation and Exhibition, if selected. Should our team fail to participate in the mid-term evaluation or the Annual Exhibition (if selected), we understand that the supported project funds will be returned to KSCST.

We also enclose the endorsement form for KSCST, Bengaluru.

Name of the students with USN No.

Signature with date

- 1.Shreya G(4JN21CS153)
- 2.Shambhavi S V(4JN22CS415)
- 3.Umme Hani(4JN21CS178)
- 4.Varshaa S K(4JN21CS180)

(Name & Signature of Project Guide with Seal)

Email id: ayeshasiddiqa@jnnce.ac.in
Contact No :9742153106

(Name & Signature of HOD with Seal)

Email id: hod_cse@jnnce.ac.in
Contact No.: 9449423361

ENDORSEMENT

(From College, endorsement to be taken in the institution / Department Letter head)

(To scan this page and enclose in the project proposal)

This is to certify that 1) Ms. Shreya G 2) Ms. Shambhavi S V 3) Ms. Umme Hani 4) Ms. Varshaa S K, are bonafide student(s) of Department of Computer Science And Engineering, in the degree program of our institution. If the project proposal submitted by these students under the 48th series of Student Project Programme is selected by KSCST, we will provide the requisite laboratory / Computer / infrastructure support in our college / Institution. Further we also take necessary steps to see that the project team will exhibit / demonstrate their project in the mid-term evaluation of project and in the Annual State-Level Poster Presentation and Exhibition (if selected). If the student team fails to send the completed project report or fails to attend the evaluation in mid-term evaluation of sanctioned projects or fails to attend the Annual State-Level Poster Presentation and Exhibition (if selected), the supported project amount will be returned to KSCST.

**(Name & Signature of
Project Guide with Seal)**

Emailid:ayeshasiddiga@jnnce.ac.in

Contact No:9742153106

**(Signature of HOD with
Seal)**

Emailid:hod_cse@jnnce.ac.in

Contact No.: 9449423361

**(Signature of the Principal
with Seal)**

Emailid:principal@jnnce.ac.in

Contact No.:9845748949

DETAILS OF PROCESSING FEES MADE THROUGH NEFT / UPI PAYMENT

(**Note:** Include this page in the softcopy of the student project proposal. The student team shall furnish the details in the Google Form. It is informed to the students to 1) keep ready the softcopy of the project proposal and other documents and 2) Furnish the payment made details as processing fees and 3) update the details in the Google Form on the same day of payment made to KSCST by NEFT / UPI payment).

1. TITLE OF THE PROJECT	:	“AGRIVISION:AI-ENHANCED YIELD FORECASTING AND SMART CROP SELECTION”
2. NAME OF THE TEAM LEADER	:	Umme Hani
3. EMAIL ID	:	ummehani181003@gmail.com
4. CONTACT MOBILE NO.	:	6363003776

PAYMENT MADE DETAILS

5. BANK REF. NO. / UTR NO. / UPI No. (12 digits)	:	
6. TRANSACTION ID	:	
7. NAME OF THE SENDER / ACCOUNT HOLDER and CONTACT NUMBER	:	
8. NAME OF THE BANK	:	
9. PROCESSING FEES	:	Rs. 1,180/- (Inclusive of 18% GST)
10. DATE OF PAYMENT MADE	:	
11. TIME	:	
12. MODE OF PAYMENT MADE (NEFT / UPI, PLEASE SPECIFY)	:	

(Name & Signature of
the team leader)

(Name & Signature of
Project Guide or HOD with Seal)

KARNATAKA STATE COUNCIL FOR SCIENCE AND TECHNOLOGY

Indian Institute of Science campus, Bengaluru

48th SERIES OF STUDENT PROJECT PROGRAMME (SPP)

(Note: This page is for information about bank details of KSCST to the student team and college / institution and not to include this page in the project proposal softcopy)

BANK ACCOUNT DETAILS OF KSCST

Name and address of the Institution	Karnataka State Council for Science and Technology, IISc Campus, Bangalore -560012
Account holder's name / Designation	Secretary, Karnataka State Council for Science and Technology
Bank Account No. & Name of the bank	Current A/C No. 0683201000024 Canara Bank, IISc Campus Branch, Bangalore-560012
IFSC Code	CNRB0000683
MICR Code	560015023
Bank Branch Address	Canara Bank, Indian Institute of Science, Bangalore-560012

BANK DETAILS

Name of the Agency	Karnataka State Council for Science and Technology IISc Campus, Bangalore - 560012
Account holder's name / Designation	Secretary , Karnataka State Council for Science and Technology
Bank Account No. & Name of the bank	Current A/C No. 0683201000024 Canara Bank IISc Campus Branch Bangalore-560012
IFSC Code	CNRB0000683
MICR Code	560015023
Bank Branch Address	Canara Bank Indian Institute of Science Bangalore-560012



केनरा बैंक Canara Bank

भारत सरकार का उपक्रम

A Government of India Undertaking



KARNATAKA STATE COUNCIL FOR SCIENCE



0683201000024@cnrb



भारत 2023 INDIA

वसुधैव कुटुम्बकम्

ONE EARTH • ONE FAMILY • ONE FUTURE